

2. Construct LALR parser for the given grammar

$$S \rightarrow L = R \mid R$$

$$L \rightarrow * R \mid id$$

$$R \rightarrow L$$

Solution

Refer the CLR for the augmented grammar, $closure(I)$ and canonical LR(1) Items.

→ check the Item contain same core elements and combine them.

$goto(I_0, *)$

$$I_4: \left\{ \begin{array}{l} L \rightarrow *.R, =/\$ \\ R \rightarrow .L, =/\$ \\ L \rightarrow .*R, =/\$ \\ L \rightarrow .id, =/\$ \end{array} \right\}$$

$$goto(I_0, *) \& goto(I_6, *)$$

$goto(I_6, *)$

$$I_{11}: \left\{ \begin{array}{l} L \rightarrow *.R, \$ \\ R \rightarrow .L, \$ \\ L \rightarrow .*R, \$ \\ L \rightarrow .id, \$ \end{array} \right\}$$

$$I_{411}: \left\{ \begin{array}{l} L \rightarrow *.R, =/\$ \\ R \rightarrow .L, =/\$ \\ L \rightarrow .*R, =/\$ \\ L \rightarrow .id, =/\$ \end{array} \right\}$$

$goto(I_0, id)$

$$I_5: L \rightarrow id., =/\$$$

$$I_{12}: L \rightarrow id., \$$$

$$goto(I_0, id) \& goto(I_6, id)$$

$$I_{512}: L \rightarrow id., =/\$$$

$goto(I_4, R)$

$$I_7: L \rightarrow *R., =/\$$$

$$I_{13}: L \rightarrow *R., \$$$

$$goto(I_4, R) \& goto(I_{11}, R)$$

$$I_{713}: L \rightarrow *R., =/\$$$

$goto(I_4, L)$

$$I_8: R \rightarrow L., =/\$$$

$$I_{10}: R \rightarrow L., \$$$

$$goto(I_4, L) \& goto(I_6, L)$$

$$I_{810}: R \rightarrow L., =/\$$$

LALR Parsing Table

State	ACTION				Go To		
	<i>id</i>	=	*	\$	S	L	R
<i>P₀</i>	S ₅₁₂		S ₄₁₁		1	2	3
<i>P₁</i>				Accept			
<i>P₂</i>		S ₆		r ₅			
<i>P₃</i>				r ₂			
<i>P₄₁₁</i>	S ₅₁₂		S ₄₁₁			810	713
<i>P₅₁₂</i>		r ₄		r ₄			
<i>P₆</i>	S ₅₁₂		S ₄₁₁			810	9
<i>P₇₁₃</i>		r ₃		r ₃			
<i>P₈₁₀</i>		r ₅		r ₅			
<i>P₉</i>				r ₁			
<i>P₁₀</i>							

1. $S \rightarrow L = R$

2. $S \rightarrow R$

3. $L \rightarrow * R$

4. $L \rightarrow id$

5. $R \rightarrow L$